

Placeholder University
Placeholder Doctoral School
Information Sciences & Technologies

Doctoral Thesis

Presented by:
Fourth STUDENT

A PLACEHOLDER TITLE FOR A PLACEHOLDER THESIS

Defended publicly on **September 30th, 2026** before the jury composed of:

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TABLE OF CONTENTS



Acknowledgements		3
I Introduction		5
I.1 Context		6
I.2 Problem Statement		6
I.3 Outline		6
II State of the Art		7
II.1 Early Approaches		8
II.2 Modern Approaches		8
II.3 Summary		8
III A Placeholder Contribution		9
III.1 Method		10
III.2 Results		10
III.3 Discussion		11
IV Conclusion & Perspectives		13
IV.1 Summary		14
IV.2 Perspectives		14
Appendices		15
A Proofs		15
A.1 Existence of a Placeholder Minimizer		16
A.2 A Proof Without a Statement		16
Bibliography		17
Acronyms		19

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This chapter carries no number, and does not appear in the mini table of contents of the chapters that follow. It can still be referred to like any other one: `\nameref{phd_acknowledgements}` gives [Acknowledgements](#).

CHAPTER I

INTRODUCTION



A PhD manuscript uses exactly the same commands as an HDR one: only the `phd` option of the document class differs.

This chapter introduces the placeholder problem addressed in this thesis, and outlines the rest of the document.

I.1	Context		6
I.2	Problem Statement		6
I.3	Outline		6

I.1 Context

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The rest of this document uses ML vocabulary, and in particular the Convolutional Neural Network (CNN). Note that CNNs is a separate entry of `acronyms.tex`, whose only purpose is to provide a plural tooltip: its full name is Convolutional Neural Networks, and it does not appear in the list of acronyms. Recall that State Of The Art (SOTA) results in this placeholder field are due to [AMET *et al.*, 2017 .

I.2 Problem Statement

Problem 1 Placeholder problem

Given a set of placeholder observations $\{\mathbf{x}_i\}_{i=1}^N$ and their placeholder targets $\{\mathbf{y}_i\}_{i=1}^N$, find a placeholder model f_θ that generalizes to unseen placeholder observations.

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. [Problem 1](#) is addressed in [Chapter III](#), after the review of [Chapter II](#).

I.3 Outline

[Chapter II](#) reviews the placeholder literature. [Chapter III](#) presents our contribution, evaluated in [Section III.2](#). [Chapter IV](#) concludes, and [Appendix A](#) gathers the proofs. The document opens on the [Acknowledgements](#) and closes on the [Bibliography](#) and the list of [Acronyms](#): these three chapters are unnumbered, and are therefore recalled by their title rather than by a number.

CHAPTER II

STATE OF THE ART



This chapter reviews the placeholder literature, and positions the contribution of [Chapter III](#) with respect to it.

II.1	Early Approaches		8
II.2	Modern Approaches		8
II.3	Summary		8

II.1 Early Approaches

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Definition 1 Placeholder representation

A *placeholder representation* of an observation x is any vector $z = g(x)$ such that z carries no more information than x .

II.2 Modern Approaches

Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat [AMET *et al.*, 2017 

```

graph LR
    Input[Input] --> Model[Model]
    Model --> Output[Output]
  
```

Figure 1 An overview of the placeholder pipeline shared by modern approaches. The same figure is reused in Chapter III, which is perfectly fine as long as the labels differ.

II.3 Summary

Table 1 compares the approaches discussed above. None of them addresses Problem 1 directly, which motivates the contribution of Chapter III.

Approach	Placeholder	Scalable	Interpretable
[IPSUM <i>et al.</i> , 1998			

CHAPTER III

A PLACEHOLDER CONTRIBUTION



This chapter presents the contribution of this thesis, and is the one that demonstrates most of the environments provided by the template.

It answers [Problem 1](#), stated in [Chapter I](#).

III.1	Method		10
III.2	Results		10
III.3	Discussion		11

III.1 Method

We look for a placeholder model f_θ minimizing the objective of Equation 1, where $\lambda > 0$ controls the strength of the placeholder regularizer.

$$\mathcal{L}(\theta) = \underbrace{\frac{1}{N} \sum_{i=1}^N \|f_\theta(\mathbf{x}_i) - \mathbf{y}_i\|_2^2}_{\text{placeholder fidelity}} + \lambda \underbrace{\|\theta\|_2^2}_{\text{placeholder regularizer}} \quad (1)$$

Proposition 1 Existence of a placeholder minimizer

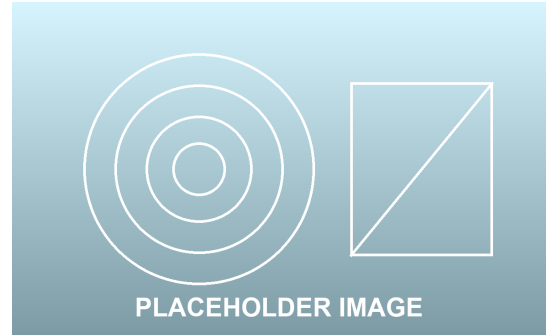
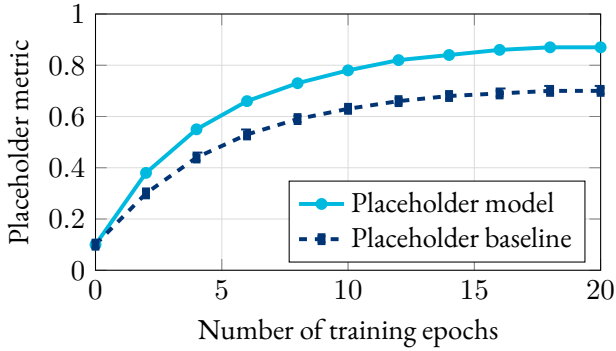
For every $\lambda > 0$, the objective of Equation 1 admits at least one minimizer θ^* .

The proof of Proposition 1 is deferred to Appendix A, so as not to interrupt the exposition. An unnumbered equation is obtained with the starred variant, which is convenient for intermediate steps:

$$\nabla_\theta \mathcal{L}(\theta) = \frac{2}{N} \sum_{i=1}^N (f_\theta(\mathbf{x}_i) - \mathbf{y}_i) \nabla_\theta f_\theta(\mathbf{x}_i) + 2\lambda\theta$$

III.2 Results

We evaluate the method of Section III.1 on placeholder datasets. Figure 2 reports the learning curves, and Table 2 the final scores.



(b) A qualitative placeholder example.

(a) Placeholder metric along training, for our model and the baseline of [AMET et al., 2017].

Figure 2 Placeholder results. (a) reports the metric along training, while (b) shows a qualitative example.

Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum. These results were partly published in [LASTNAME et al., 2021] and [COAUTHOR et al., 2024].

Dataset	[AMET <i>et al.</i> , 2017 	Ours
Placeholder-A	0.70 ± 0.02	0.87 ± 0.01
Placeholder-B	0.61 ± 0.03	0.79 ± 0.02
Placeholder-C	0.55 ± 0.04	0.68 ± 0.03

Table 2 Final placeholder scores, averaged over five placeholder runs. Best results in bold.

III.3 Discussion

 Limitations

An unnumbered box, obtained with `boxenv*`, is handy for remarks that do not need to be cross-referenced. The placeholder method has placeholder limitations, which are discussed in [Chapter IV](#).

CHAPTER IV

CONCLUSION & PERSPECTIVES



A short closing chapter, summarizing the contribution of [Chapter III](#) and listing the directions it opens.

IV.1	Summary	14
IV.2	Perspectives	14

IV.1 Summary

This thesis addressed [Problem 1](#), introduced in [Chapter I](#). After the review of [Chapter II](#), [Chapter III](#) proposed a placeholder method, whose objective is given in [Equation 1](#) and whose results are reported in [Table 2](#).

IV.2 Perspectives

- Extending the placeholder method to placeholder data of higher dimension.
- Reducing the GPU cost of placeholder training, so that the approach scales beyond placeholder datasets.
- Studying the interpretability of the placeholder representations of [Definition 1](#).

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APPENDIX A

PROOFS



Deferring proofs to an appendix keeps the main text readable.

The `proof` environment takes the label of the statement it proves, and displays a clickable reference to it in its title.

A.1	Existence of a Placeholder Minimizer	16
A.2	A Proof Without a Statement	16

A.1 Existence of a Placeholder Minimizer

Proof of Proposition 1

The objective of Equation 1 is continuous in θ , and coercive as soon as $\lambda > 0$, since $\|\theta\|_2^2 \rightarrow \infty$ when $\|\theta\|_2 \rightarrow \infty$. A continuous coercive function on $\mathbb{R}^d$ attains its infimum, hence the existence of a minimizer θ^* . $\square$

A.2 A Proof Without a Statement

The mandatory argument of proof can also be left empty, in which case the box is simply titled “Proof”.

Proof

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ACRONYMS



AI		Artificial Intelligence
BSc		Bachelor of Science
CNN		Convolutional Neural Network
GNU		GNU's Not Unix
GPL		General Public License
GPU		Graphics Processing Unit
HDR		Habilitation à Diriger des Recherches
ML		Machine Learning
MSc		Master of Science
PDF		Portable Document Format
PhD		Philosophiæ Doctor
SOTA		State Of The Art

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